



RAIN DETECTOR WITH WARDROBE SHELTER

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Aim:

To design and implement an automated system that detects rainfall and subsequently activates a mechanism to shield clothes in a wardrobe from getting wet.

Introduction:

The Rain Detector with Wardrobe Shelter is an automated system that protects clothes from getting wet during unexpected rain. Using a rain sensor, Arduino, and servo motor, the system detects rain and automatically covers the clothes with a shelter. It offers a smart and convenient solution for drying clothes safely, especially in unpredictable weather.

Component Used:

Microcontroller: Arduino Uno

- Acts as the brain of the system. It takes input from the Rain Sensor Module. Processes the data to check if it's raining. Based on that, it sends signals to the Servo Motor to either open or close the shelter. Controls timing, decision-making, and signal flow between all connected components.

Rain Sensor Module:

- Detects the presence of rain. It has a rain detection board that senses water droplets. When rain touches the sensor, it changes the electrical resistance. The sensor sends a digital (rain detected) or analog (how much rain) signal to the Arduino.

Servo Motor:

- Drives the movement of the shelter mechanism.

Power Supply: 12V DC adapter.

Miscellaneous: Wires, connectors, and structural materials for the shelter.

Circuit Connections:

- Connect the rain sensor module's output to an analog input pin on the Arduino.
- Attach the servo motor's control wire to a PWM-capable digital pin on the Arduino.
- Ensure the power supply is appropriately connected to both the Arduino and the servo motor, considering their voltage requirements.

● Soil Moisture Sensor:

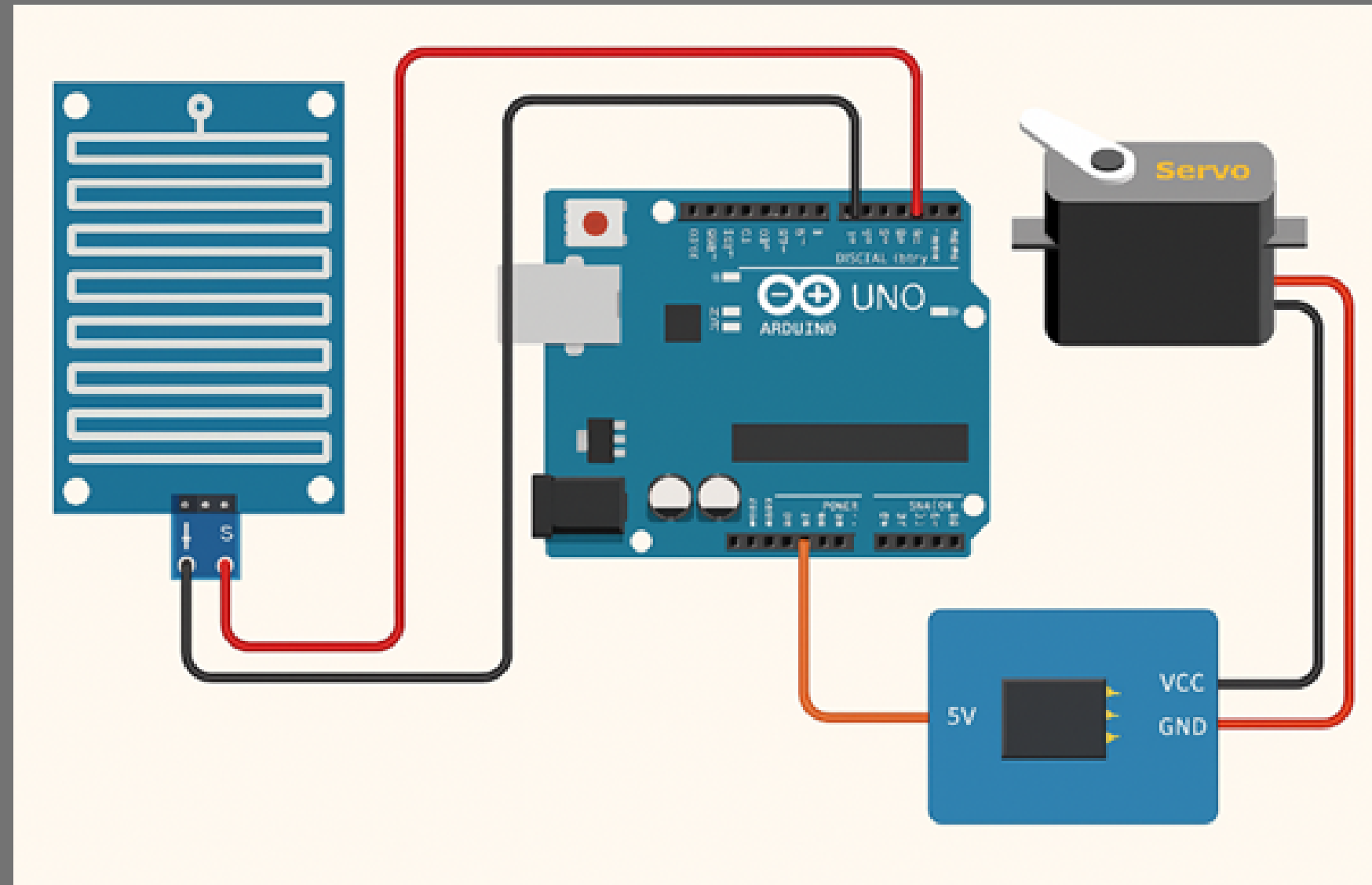
- VCC → Arduino 5V
- GND → Arduino GND
- Signal (A0) → Arduino Analog Pin A0

● Servo Motor:

- VCC → Arduino 5V
- GND → Arduino GND
- Signal → Arduino Digital PWM Pin (e.g., D9)

Simulation:

Utilize simulation software like Proteus to model the circuit. This allows for testing the system's response to simulated rainfall before actual implementation.

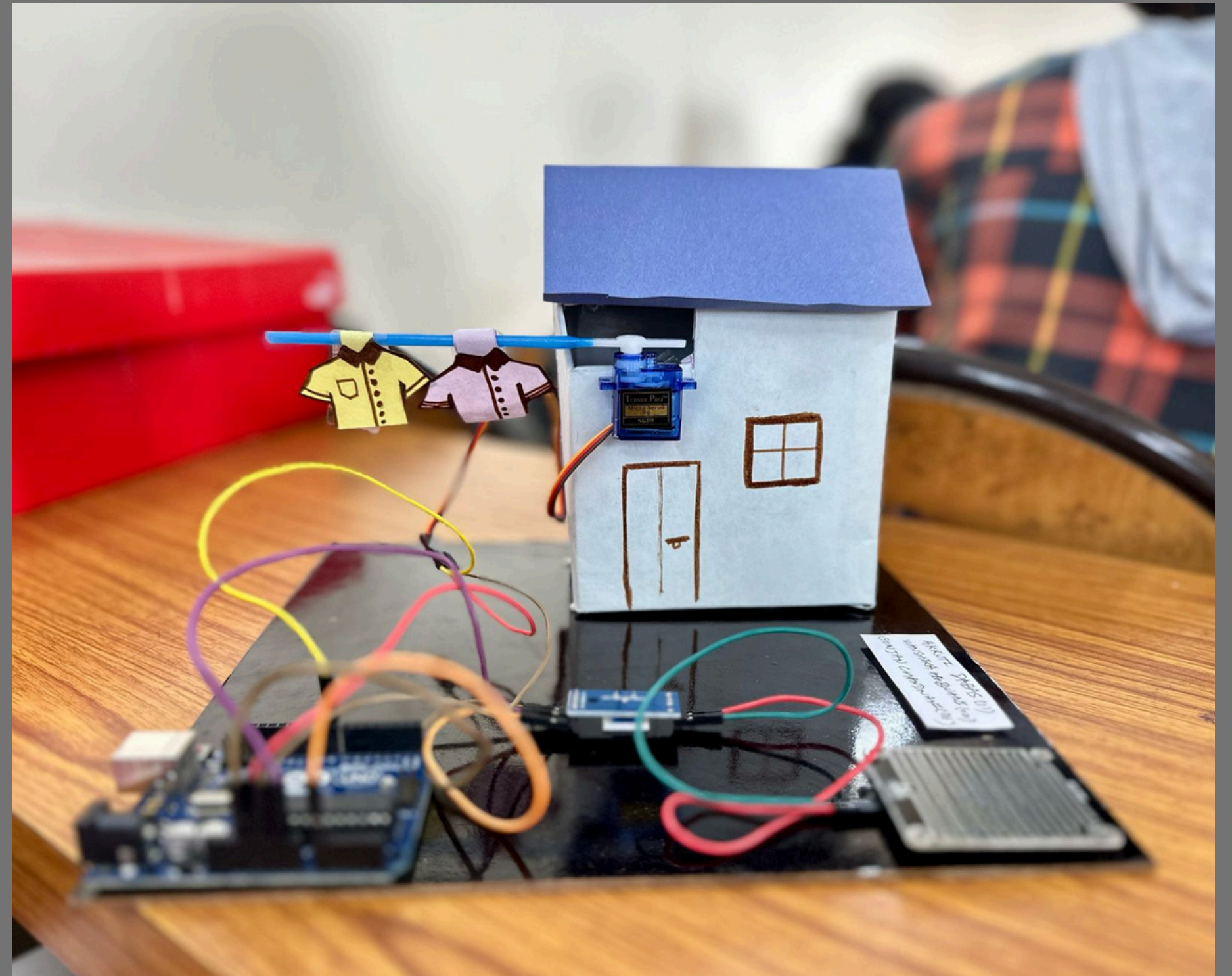


Working Principle:

- The rain sensor detects moisture and sends a signal to the Arduino.
- Upon receiving this signal, the Arduino actuates the servo motor.
- The servo motor then moves the shelter to cover or uncover the clothes based on the presence or absence of rain

Circuit Diagram and Working:

- Include a schematic diagram illustrating the connections between the Arduino, rain sensor, and servo motor.
- Explain the flow: when rain is detected, the sensor's signal prompts the Arduino to activate the servo motor, deploying the shelter over the clothes. Once the rain stops, the shelter retracts.



Arduino Algorithm:

- Initialize the input and output pins.
- Continuously monitor the rain sensor's output.
- If rain is detected (sensor output exceeds a predefined threshold), send a command to the servo motor to deploy the shelter.
- If no rain is detected, command the servo motor to retract the shelter.

Application:

- Primarily used in households to protect clothes from unexpected rain.
- Can be adapted for use in outdoor markets or events to protect goods from rain.

Conclusion:

- The project effectively demonstrates an automated solution to protect clothes from rain using readily available components.
- It showcases the practical application of integrating sensors and actuators with microcontrollers to solve everyday problems.